

Module 12: TypeScript in Real-World Applications

1. Introduction to Real-World TypeScript Applications

TypeScript is widely used in web development, backend services, mobile apps, and enterprise applications.

2. TypeScript with Frontend Frameworks

TypeScript with React

Install TypeScript in a React project:

```
sh
```

```
npx create-react-app my-app --template typescript
```

Component with TypeScript:

```
tsx
```

```
import React from "react";
```

```
interface ButtonProps {
```

```
  label: string;
```

```
  onClick: () => void;
```

```
}
```

```
const Button: React.FC<ButtonProps> = ({ label, onClick }) => {
```

```
  return <button onClick={onClick}>{label}</button>;
```

```
};
```

```
export default Button;
```

3. TypeScript in Backend Development

Using TypeScript with Node.js and Express

Install TypeScript and Express:

```
sh
```

```
npm install express typescript @types/express
```

Basic Express server with TypeScript:

```
typescript
```

```
import express, { Request, Response } from "express";
```

```
const app = express();
```

```
const port = 3000;
```

```
app.get("/", (req: Request, res: Response) => {  
  res.send("Hello, TypeScript with Express!");  
});
```

```
app.listen(port, () => {  
  console.log(`Server running on port ${port}`);  
});
```

4. TypeScript in Mobile App Development

Using TypeScript with React Native

```
sh
```

```
npx react-native init MyApp --template react-native-template-typescript
```

5. TypeScript in Enterprise Applications

- Used in large-scale projects for safety and maintainability.
- Supports microservices architecture.

6. Best Practices for TypeScript in Production

- Enable `strict` mode in `tsconfig.json`.
- Use interfaces for better maintainability.
- Implement logging and error handling.

Practice Questions:

1. How do you set up a TypeScript React project?
2. Write a simple Express API using TypeScript.
3. How does TypeScript improve React component development?
4. Implement a basic microservice in TypeScript.

5. Why is TypeScript beneficial in enterprise applications?

Practice Answers:

1. Use ``npx create-react-app my-app --template typescript``.

2. Simple Express API example:

`typescript`

```
import express, { Request, Response } from "express";
```

```
const app = express();
```

```
app.get("/api", (req: Request, res: Response) => res.json({ message: "API working!" }));
```

```
app.listen(3000, () => console.log("Server running on port 3000"));
```

3. TypeScript enforces prop types, preventing errors at runtime.

4. Basic microservice example:

`typescript`

```
import express from "express";
```

```
const app = express();
```

```
app.get("/health", (req, res) => res.json({ status: "OK" }));
```

```
app.listen(5000, () => console.log("Microservice running on port 5000"));
```

5. TypeScript improves code quality, reduces bugs, and enhances maintainability.